

SRM University AP
Practice Sheet 4
Unit 4

1. Are the following sets of functions defined on the given interval I linearly dependent or linearly independent? Justify your answer.

- (a) $f_1(x) = 1$, $f_2(x) = x$, $f_3(x) = x^3$ on $(-\infty, \infty)$.
- (b) $f_1(x) = e^x$, $f_2(x) = 2e^x - 3e^{4x}$ on $(-\infty, \infty)$.
- (c) $f_1(x) = e^x$, $f_2(x) = e^{4x}$, $f_3(x) = 2e^x - 3e^{4x}$ on $(-\infty, \infty)$.
- (d) $f_1(x) = e^{-x} \sin x$, $f_2(x) = e^{-x} \cos x$ on $(-\infty, \infty)$.
- (e) $f_1(x) = x^2$, $f_2(x) = x|x|$ on $(0, \infty)$.
- (f) $f_1(x) = x^2$, $f_2(x) = x|x|$ on $(-\infty, \infty)$.

2. If the Wronskian W of f_1 and f_2 is x^2e^x and $f_1(x) = x$, find $f_2(x)$.

3. In each of the following, verify that y_1 and y_2 are linearly independent solutions of the given ODE on $[0, 2]$, and find the particular solution satisfying the given initial conditions:

- (a) $y'' + y' - 2y = 0$, $y_1 = e^x$, $y_2 = e^{-2x}$, $y(0) = 8$, $y'(0) = 2$.
- (b) $y'' + 5y' + 6y = 0$, $y_1 = e^{-2x}$, $y_2 = e^{-3x}$, $y(0) = 1$, $y'(0) = 1$.
- (c) $y'' + y' = 0$, $y_1 = 1$, $y_2 = e^{-x}$, $y(2) = 0$, $y'(2) = e^{-2}$.
- (d) $y'' + 2y' + 2y = 0$, $y_1 = e^{-x} \sin x$, $y_2 = e^{-x} \cos x$, $y(0) = 0$, $y'(0) = 15$.

4. Solve the following linear homogeneous ODEs with constant coefficients:

- (a) $y'' - 3y' + 2y = 0$
- (b) $y'' - 6y' + 9y = 0$
- (c) $y'' - 6y' + 25y = 0$
- (d) $y'' - 2y' + 3y = 0$
- (e) $y'' - 2y' = 0$
- (f) $y'' = 0$

5. A second-order differential equation and a function y_1 are given. Verify that y_1 satisfies the equation and find the general solution using reduction of order:

- (a) $(1 - x^2)y'' - 2xy' + 2y = 0$, $y_1 = x$
- (b) $(x^2 + 1)y'' - 2xy' + 2y = 0$, $y_1 = x$
- (c) $x^2y'' + xy' + (x^2 - \frac{1}{4})y = 0$, $y_1 = \frac{\sin x}{\sqrt{x}}$, $x > 0$
- (d) $xy'' - (2x + 1)y' + (x + 1)y = 0$, $y_1 = e^x$
- (e) $x^2y'' + xy' - 4y = 0$, $y_1 = x^2$

6. Solve the following using variation of parameters:

(a) $y'' - 2y' - 3y = -3e^{-x}$

(b) $y'' + y = 3\sin(2x) + \cos(2x)$

(c) $y'' + 2y' + 5y = 4e^{-x}\cos(2x)$, $y(0) = 1$, $y'(0) = 0$

(d) $y'' + 4y = \sin(2x)$

(e) $y'' + 4y = \tan(2x)$

(f) $y'' + 2y' + y = e^{-x}\ln x$

(g) $(x^2 + 1)y'' - 2xy' + 2y = 6(x^2 + 1)^2$, given $y_1 = x$, $y_2 = x^2 - 1$

(h) $x^2y'' - 6xy' + 10y = 3x^4 + 6x^3$, given $y_1 = x^2$, $y_2 = x^5$